

1. Administrative & Study Identification

Official Study Title: A Phase III, Multicenter, Single-Arm Study Evaluating the Efficacy, Safety, Pharmacokinetics, and Pharmacodynamics of Crovalimab in Adult and Adolescent Patients With Atypical Hemolytic Uremic Syndrome (aHUS) **Brief Study Title:** A Study Evaluating the Efficacy, Safety, Pharmacokinetics and Pharmacodynamics of Crovalimab in Adult and Adolescent Participants With Atypical Hemolytic Uremic Syndrome (aHUS) **Short Title/Acronym:** COMMUTE-a Trial **Protocol Number:** B042353 **NCT Number:** NCT04861259 **Trial Status:** ACTIVE_NOT_RECRUITING **Sponsor/ Company Name:** Hoffmann-La Roche (Roche) **Investigational Product:** Crovalimab **Phase of Development:** Phase III **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy and safety of crovalimab in adult and adolescent participants with atypical hemolytic uremic syndrome (aHUS). **Secondary Objectives:** To evaluate the pharmacokinetics (PK) and pharmacodynamics (PD) of crovalimab, alongside changes in renal function and dialysis status.

2.2 Background & Rationale To address the treatment burden for patients with aHUS. Existing Complement Component 5 (C5) inhibitors are effective for aHUS management but typically require regular and burdensome intravenous (IV) infusions every two to eight weeks. Crovalimab is a novel C5 inhibitor with low-volume, subcutaneous (SC) administration every four weeks. This study evaluates its efficacy and safety to provide a more convenient, self-administered treatment option that protects red blood cells, reduces disease symptoms, and improves patients' quality of life.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Participants are assigned to three cohorts: Naive Cohort, Switch Cohort, and C5 SNP Cohort) **Blinding:** Open-label **Randomization:** N/A (Non-randomized assignment based on prior C5 inhibitor treatment history and genetics)

3.2 Planned Enrollment & Duration Targeted Enrollment: 83 participants **Study Population:** Adult and adolescent participants (\$\ge\$ 12\$ years, body weight \$\ge\$ 40\$ kg) **Number of Sites:**

Multicenter (Global, including sites in the US, EU, Brazil, and India) **Estimated Study Start:** 2021 **Estimated Primary Completion:** August 2029

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years for adults, or adolescents ≥ 12 years old; body weight ≥ 40 kg at screening. 2. Confirmed diagnosis of atypical hemolytic uremic syndrome (aHUS). 3. Vaccination against *Neisseria meningitidis* serotypes A, C, W, and Y, and vaccination against serotype B, according to national recommendations. 4. Vaccinated against *Haemophilus influenzae* type B and *Streptococcus pneumoniae*, according to national vaccination recommendations. 5. Female participants of childbearing potential must have a negative serum pregnancy test result within 7 days prior to initiation of crovalimab and agree to remain abstinent or use contraception. 6. Patients receiving concurrent therapies (e.g., immunosuppressants, mTORi, corticosteroids) must be on a stable dose for ≥ 28 days prior to screening.

4.2 Exclusion Criteria 1. Pregnant or breastfeeding or intending to become pregnant. 2. Thrombotic microangiopathy (TMA) associated with non-aHUS related renal disease, identified drug exposure-related TMA, or a positive direct Coombs test. 3. Chronic dialysis within 90 days prior to first crovalimab administration and/or end-stage renal disease. 4. History of *Neisseria meningitidis* infection within 6 months of study enrollment, or active systemic bacterial, viral, or fungal infection within 14 days before the first crovalimab dose. 5. Positive Human Immunodeficiency Virus (HIV) test. 6. Presence of fever ($\geq 38^{\circ}\text{C}$) or multi-system organ dysfunction/failure.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Initial weight-based loading dose administered intravenously (IV), followed by 340 mg administered subcutaneously (SC) on Week 1 Day 2, and Weeks 2, 3, and 4. From Week 5 and every 4 weeks (Q4W) thereafter, a maintenance dose of 680 mg SC (for body weight 40 to <100 kg) or 1020 mg SC (for body weight ≥ 100 kg). **Route of Administration:** Intravenous (IV) loading followed by Subcutaneous (SC) injections. **Duration of Treatment:** Long-term continuous maintenance (through expected trial completion in August 2029).

5.2 Concomitant & Prohibited Therapy Permitted: Stable doses (for ≥ 28 days prior to screening) of immunosuppressants, corticosteroids, mammalian target of rapamycin inhibitors (mTORi), or calcineurin inhibitors. **Prohibited:** Recent

intravenous immunoglobulin (IVIg) treatment, recent use of tranexamic acid, and participation in another interventional treatment study with an investigational agent within 28 days of screening.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Pregnancy Test, Vaccination Verification
Baseline	Day 1	First Dose (IV loading), Efficacy & Safety Baseline, Quality Audit
Interim	Weeks 1-4	SC Loading Doses (340 mg), PK/PD Labs, Efficacy Assessment, Education on SC administration
End of Study	Week 5+ (Q4W)	Maintenance SC Dosing (680/1020 mg), Safety tracking, Exit Interview, IP Accountability

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Percentage of Participants with Complete Thrombotic Microangiopathy Response (cTMAr). **Measurement Method:** Central Laboratory Analysis (platelet counts, LDH, eGFR) and regular clinical assessments.

7.2 Secondary & Exploratory Endpoints 1. Change from Baseline in Dialysis Status. 2. Change from Baseline in Estimated Glomerular Filtration Rate (eGFR). 3. Percentage of Participants with Change from Baseline in Chronic Kidney Disease (CKD) Stage. 4. Pharmacokinetics (PK) of crovalimab (serum drug concentration). 5. Pharmacodynamics (PD) of crovalimab (Complement C5 inhibition parameters).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via onsite visits and patient monitoring, with rigorous tracking for signs of *Neisseria meningitidis* or other severe systemic infections. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Internal/External safety monitoring mechanisms managed by Hoffmann-La Roche.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for efficacy outcomes across the three distinct cohorts (Naive, Switch, and C5 SNP). Safety Set for all AE/SAE evaluations. **Sample Size Justification:** Appropriately powered to evaluate the clinical improvement in TMA parameters and non-inferiority/

superiority in the ultra-rare aHUS patient population ($\alpha = 0.05$, $\text{Power} = 80\%$). Target enrollment is set to 83. **Handling of Missing Data:** Multiple Imputation (MI) and Last Observation Carried Forward (LOCF) for missing continuous laboratory endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite monitoring during the loading phase and remote monitoring tracking adherence to the self-administered SC regimen. **Audit Trail:** Details on eCRF locking and data clarification forms via standard quality audit mechanisms.

11. Study Identifiers & Governance

Primary ID: {{B042353}} **Registry Number:** {{NCT04861259}} **Sponsor/Lead Organization:** {{Hoffmann-La Roche (Roche)}} **Study Title:** {{COMMUTE-a Trial}} **Official Regulatory Title:** {{A Phase III, Multicenter, Single-Arm Study Evaluating the Efficacy, Safety, Pharmacokinetics, and Pharmacodynamics of Crovalimab in Adult and Adolescent Patients With Atypical Hemolytic Uremic Syndrome (aHUS)}}

12. Clinical Framework (aHUS Specific)

Primary Condition: {{Atypical Hemolytic Uremic Syndrome (aHUS)}} **Diagnosis Threshold:** {{Evidence of Thrombotic Microangiopathy (TMA)}} **Medical Methodology:** {{Complement Component 5 (C5) Inhibition}} **Optimization Target:** {{Complete TMA response and stable kidney function}}

13. Study Procedures & Methodology

Management Pathway: {{Transition to SC self-administered C5 inhibitor}} **Education Protocol (HCP):** {{Training on SC administration and meningococcal infection monitoring}} **Education Protocol (Patient):** {{SC self-injection training and infection risk awareness}} **Quality Audit Mechanism:** {{Patient adherence monitoring to SC regimen and safety tracking}}

14. Observation & Follow-Up Schedule

Total Duration: {{Follow-up through August 2029}} **Onsite Visit Cadence:** {{Loading phase visits, then Q4W matching SC dosing}}

Remote Monitoring Frequency: {{Telehealth/remote monitoring as required}} **Checklist Review Interval:** {{Every 12 weeks (Q12W)}}

15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Name]	_____	[07-Apr-2026]					
Clinical Operations Lead	[Name]	_____	[07-Apr-2026]					
Statistician	[Name]	_____	[07-Apr-2026]					